

MAINUDDIN TALUKDAR

AI Engineer | Full-Stack (TypeScript · Next.js · Python) | Master of AI Student, University of Canterbury

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PROFILE

Software engineer and Master of Artificial Intelligence student with nine years' experience designing, building and shipping production systems, now focused on applied AI. I build LLM-powered agents end to end, using Claude and Claude Code, Cursor and Codex well beyond chat — for agentic coding, tool use and automated workflows — and combine that with full-stack delivery in TypeScript, Next.js and Python. My side projects put AI agents into real New Zealand small-business workflows, so I understand both the engineering and how these tools earn their place in a business. Self-directed, comfortable owning a project end to end, and keen to help benchmark and ship agents for NZ SMEs.

KEY SKILLS

- **AI Agents & LLMs:** Claude and Claude Code, Cursor, Codex used for agentic coding and tool use (not just chat); LLM integration, agent architectures, prompting strategies, RAG, benchmarking and evaluation.
- **Full-Stack Web:** TypeScript/JavaScript, Next.js, React, React Native, Node.js, REST API design, Supabase and PostgreSQL, Vercel deployment, functional component design.
- **Python:** Python for scripting, data pipelines, ML and deep learning, automation and backend services; NumPy/Pandas and PyTorch-style workflows.
- **Cloud & Delivery:** AWS (Lambda, DynamoDB, S3), Docker, Redis, CI/CD, analytics dashboards; requirements analysis, design, build, unit/integration testing, deployment and documentation.
- **Ways of Working:** Self-directed, end-to-end ownership of features, cross-functional and client-facing collaboration, clear written and verbal communication.

PROFESSIONAL EXPERIENCE

Software Development Engineer II — Aperia Solutions (Client: Fiserv)

Aperia Solutions, New Jersey, USA • Aug 2024 – Feb 2026

- Provided technical leadership on assigned features, owning requirements analysis, design, build, test, deployment and documentation across payment and reporting services.
- Built and optimised RESTful APIs and enterprise integrations, improving response time by **~40% under peak load**, and integrated Redis caching to cut average read latency by over 80% for real-time updates serving millions of daily users.
- Automated build, test and deployment through CI/CD pipelines with Git workflows, Jira tracking and monitoring, and communicated technical trade-offs clearly to internal and client stakeholders.

Software Developer → Senior Software Developer — Vantage Labs (Remote, USA)

Vantage Labs, Pennsylvania, USA • Jun 2019 – Aug 2023

- Built full-stack products end to end in **TypeScript, React and Node.js** — including two React Native apps shipped to the Apple App Store and Google Play — with REST APIs, Firebase authentication and real-time data.
- Led data-analysis pipelines in Python with Hadoop and Spark to process large-scale performance data, driving athlete recommendations that lifted revenue ~9.7% annually.
- Wrote unit and integration tests, resolved UI performance issues, deployed to AWS (Lambda, S3, NoSQL) with monitoring, and was promoted to Senior for sustained impact.

Software Developer — TigerIT Bangladesh Ltd

TigerIT Bangladesh Ltd, Dhaka • Jun 2016 – Jun 2019

- Delivered national-scale government systems serving over 100 million people, built on Spring Boot, Oracle and WebLogic, cutting licence processing time by ~30%.
- Built automated test suites and reporting pipelines with SparkSQL and Kafka to improve reliability and deliver real-time operational insight over 35 TB of data.

AI PROJECTS & PORTFOLIO

- **TradiePulse (tradiepulse.mainuddintalukdar.cloud)** — A conversational AI agent that lets a customer describe a household problem in plain language, the way they would to a person. The agent interprets the request, then finds the nearest available and suitably qualified tradesperson — plumber, electrician or mechanic — and connects the two. Built with a Next.js front end, a Python agent service for language understanding and matching, and Supabase for tradie profiles, availability and jobs; containerised with Docker and served on my Hostinger VPS.
- **MathQuest (mathquest.mainuddintalukdar.cloud)** — An interactive maths learning app for primary-school children, built on Next.js, Supabase and Vercel. Children work through puzzle-based lessons while the app tracks how they engage — time per question, attempts and accuracy — in Supabase, then scores each child and suggests what to practise next. The scoring and suggestions run as a separate Python microservice, proving the end-to-end workflow ahead of a fully trained model.
- **Engineering approach** — Each project runs as its own Dockerised service, ships through a GitHub CI/CD pipeline with automated tests and smoke checks, and is served on its own subdomain via Caddy — a repeatable path from commit to production.
- **Portfolio (mainuddintalukdar.cloud)** — Full career history, company work and pet projects, with architecture notes and live demos in one place.

EDUCATION

- **Master of Artificial Intelligence** — University of Canterbury, Christchurch — current (NZQA Level 9) • Deep Learning, Robotics, Information Visualisation.
- **Master of Science in Computer Science** — Maharishi International University, Iowa, USA — 2025 • Enterprise Architecture, Cloud Computing, Big Data, Algorithms.
- **Bachelor of Science in Software Engineering** — University of Dhaka, Bangladesh — 2016.

PUBLICATION & ACHIEVEMENTS

- **IEEE Publication** — “A Stacked Meta-Model Framework for Diabetes Prediction”, accepted at ICCIT (IEEE); applied feature engineering and meta-learning to a real ML problem.
- **Champion, DhakaThon 2015** — First place among 100+ teams in a national two-day application-development hackathon.

INTERESTS

- **Sport** — Play cricket and table tennis and follow football; keeps me active and reinforces a disciplined, team-first mindset.
- **Exploring places & cultures** — Enjoy walking, hiking and travelling to new places, staying curious and adaptable through fresh perspectives.
- **Connecting with people** — Value meeting new people and building genuine, lasting professional relationships, learning from those around me.

REFEREES

- **Dr. Mrudula Mukadam** — Chair & Associate Professor of Computer Science, Maharishi International University • mmukadam@miu.edu • +1 641 233 5491.
- **Shaikh Ahasanul Haque** — Software Development Lead (Supervisor), Vantage Labs • shaque@vantage.com • +880 1819 484992.